Process: building experiments

Scientific method

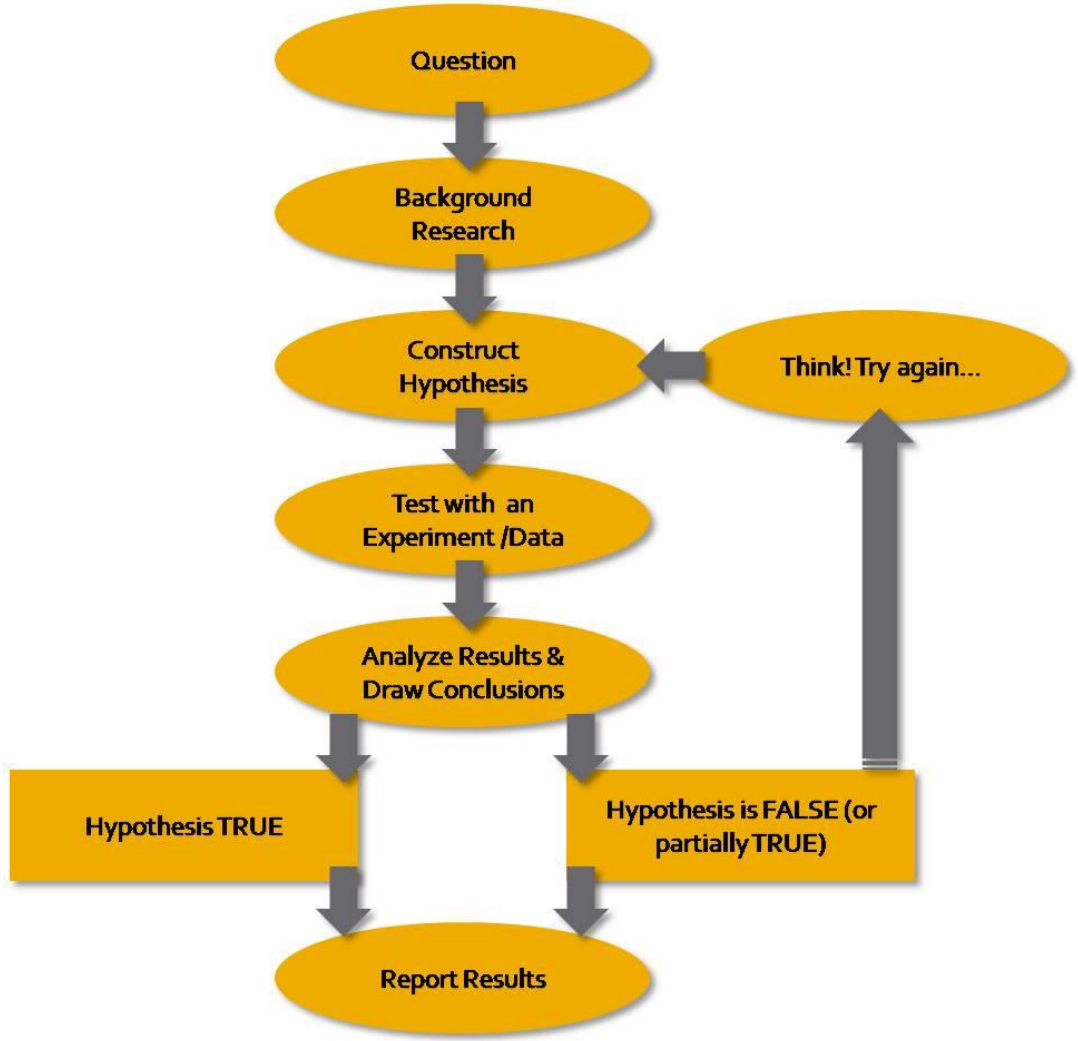
Take a logical approach to any problem.

Individual paths may vary but any effort relying on an experiment has similar steps:

- State the problem.
- Gather information about the subject.
- Form a hypothesis.
- Test the hypothesis/experiment.
- Gather results.
- Make and share the conclusions.

Apply to software and business problems.

Emphasize its criticality to Third Way resolution.



Source: https://chem.libretexts.org/Bookshelves/General_Chemistry/General_Chemistry_Supplement_(Eames)/Chemistry_Basics/Scientific_Method

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Building an experiment

Experiments emerge as part of research design to identify variables:

- **Independent variable:** the manipulated quantity.
- **Dependent variable:** responds to change.
- **Constant:** factors which could change but do not.
- **Hypothesis:** if/then statement to test during experiment. If (the independent variable) makes this planned change, then (the dependent variable) will react in this way.

Four-question experiment design

- What is available to experiment on?
- What action should be observed?
- How can I change materials to influence action?
- How can I measure the change?



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Analytic tools

Qualitative

- They are designed to handle content analysis, transcription, discourse analysis, coding (narrative), recursive abstraction.
- **CAQDAS:** computer-assisted qualitative data analysis.
- **Nvivo:** helps organize and analyze unstructured data.
- **Quirkos:** tag and code data to compare themes.
- options to fit needs.

Quantitative

- They are designed to handle high-level math functions and comparison.
- Excel: spreadsheets and embedded analysis tools.
- Statistical Package for Social Sciences (SPSS) Statistics (IBM); conduct interactive or batched statistical analysis.
- R
  - language and environment for statistical computing and graphs
  - integrated software sheet.



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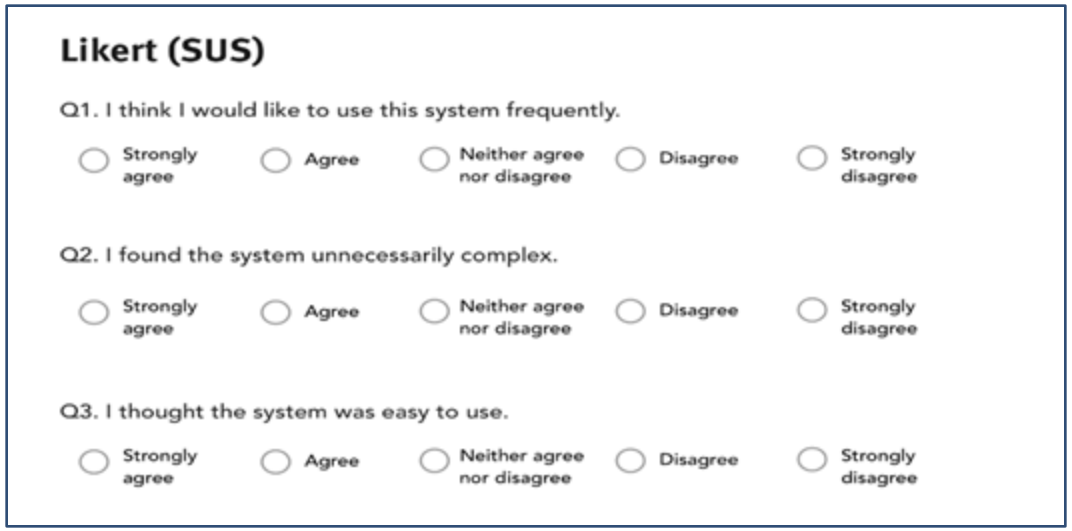
UI/UX rating scales

Likert

- rates perception of a thing based on 1-5 scale, a measure agreement:
  1. Strongly agree
  2. Agree
  3. Neither
  4. Disagree
  5. Strongly disagree
- supports User Interface/User Experience (UI/UX) interactions
- vulnerable to:
  - Acquiescence bias: tendency to agree with others.
  - Social desirability bias: report views that others see as favorable.

Semantic differential

- rates task on a bipolar adjectival scale
- based on two assumptions:
  - Adjectives are truly bipolar.
  - Respondents understand dichotomy.





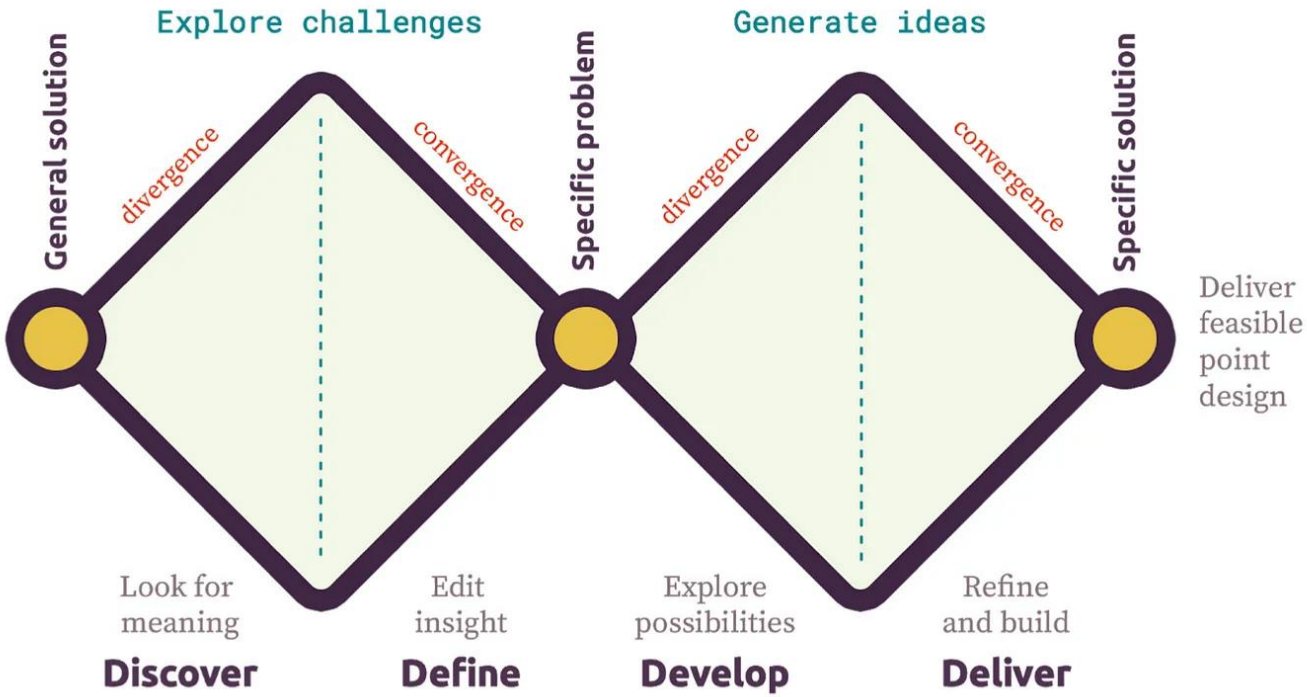
Source: https://www.nngroup.com/articles/rating-scales/

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Interviews and surveys (1/2)

Tips for rating scales in UX surveys:

- Use standardized usability questions.
- Try with Likert and Semantic on different groups.
- Use existing labels.
- Ensure true opposites.
- Use optional text fields.
- Add a not applicable option.



Source: https://betterdesigned.substack.com/p/better-designed-explains-how-to-conduct

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Interviews and surveys (2/2)

- Sometimes you need more than a numerical rating.
- Interviews learn what user is trying to do, and what works for them.

Tips for interview:

- Do not ask leading questions.
- Do not ask what they want, ask about problems.
- No Yes/No, but open-ended questions.
- Identify assumptions.
- Use the same questions, and from multiple angles.
- Follow-up.
- Use silence and ask users to show you.
- Record when possible and take notes.

Tips for surveys:

- Keep personal questions to the end.
- Do not be too long; 10-20 minutes.
- Keep answers balanced (Likert).
- Do not ask yes/no questions.
- Focus on one item (not how would you rate quality of product and support).
- Preview before you send.

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